$$\hat{X}(\tau) = \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{S}}\tau}\,\hat{X}\,\mathrm{e}^{-\mathrm{i}\hat{H}_{\mathcal{S}}\tau} = \hat{X}\cos(\Omega\tau) + \frac{1}{M\Omega}\hat{P}\sin(\Omega\tau)\,. \tag{5.24}$$